

Mott localisation, antiferromagnetism and nematic superconductivity in the extended Hubbard model with non-local pairing

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Copper oxides based compounds have been widely studied in recent decades, due to their peculiar phase diagram and the wide observed phenomenology. Among the various phases, these materials exhibit long-range antiferromagnetic ordering when undoped, and high-temperature d -wave superconductivity when charge is injected. The latter, when discovered, was particularly unexpected: indeed, at normal state these materials show an insulating phase. It has been understood, however, that the insulation mechanism is incompatible with conventional band theory; it results, instead, from the strong on-site Coulomb repulsion between electrons. Namely, these materials are Mott insulators. Both the antiferromagnetic and superconducting phases are deeply connected to the physics of Mott insulation.

Most of the low-energy properties of copper oxides are well described by the Hubbard model on bidimensional square lattice. The latter is composed of two parts: the kinetic operator, representing the hopping of electrons on neighbouring sites, and the on-site repulsion between electrons of opposite spin, which originates from Coulomb interaction. The Hubbard model can host Mott insulation, antiferromagnetism and superconductivity, and advanced numerical techniques lead in general to very good compatibility with many experiments.

In recent years, it has been proposed that a more accurate description of the low-energy spectrum of copper oxides could have been reached by adding a new interaction to the Hubbard model, namely a non-local attraction between electrons on neighbouring sites. The resulting model has been denominated “extended Hubbard model”. Among other reasons, it is well known that an interaction of this kind is compatible with the presence of anisotropic superconductivity and, moreover, could act as a source of d -wave Cooper pairing. In this framework, it has not been explored yet how this interaction possibly relates to Mott insulation.

In this thesis we study, both theoretically and numerically, the extended Hubbard model using the Hartree-Fock method. Namely, we implement a constrained variational search in a subspace of the many-body Hilbert space. In the Hartree-Fock method, the latter is identified with the set of the ground-states of all possible non-interacting fermionic hamiltonian. By constraining to such a class of states, we are able to map the variational search, which involves a minimisation procedure, to an equivalent self-consistency problem, which is described by the search for a fixed point of a non-linear map acting in an abstract parametric space. The solution of the original variational problem is expressed in terms of the coordinates of this fixed point. In this framework, the problem can be solved numerically by the means of an iterative scheme.

Despite its simplicity, the Hartree-Fock method provides a powerful framework to study competition among phases. Each competing phase is described by the spontaneous breaking of one or more symmetries of the extended Hubbard model. Hence, by constraining the variational search to the subspace of states that show the symmetries of a given phase, we can obtain the best approximation to the true ground-state inside that subspace. Then, comparing the free energies of the resulting approximations, we can assess which phase of those studied is energetically favoured. We study competition among three phases: normal, antiferromagnetic and superconducting. For each, we derive the analytic expression of the non-linear map that defines the self-consistency problem, we solve it numerically and discuss the results. We run simulations at approximately zero temperature, varying the strength of the local repulsion and the non-local attraction, and by doping the material through injection of charge.

For the normal phase, we find theoretical constraints on the Hartree-Fock parametric solution indicating that the non-local attraction acts as a source of Mott localisation, namely by renormalising the hopping amplitude. As a consequence, the bandwidth is reduced. Numerical results confirm theoretical expectations, and a region of complete Mott localisation is found in parametric space, delimited by a boundary which we were able to approximate analytically. Inside this region we observe a total flattening of the bare electronic bands. It is worth mentioning that for the

conventional Hubbard model, described by the on-site repulsion alone, Mott localisation is not found at Hartree-Fock level.

We study long-range antiferromagnetic ordering limiting to null electronic doping, based on theoretical stability considerations. To start, we study the insurgence of antiferromagnetism in the conventional Hubbard model. Then we add the non-local pairing and highlight the key differences. In this context, similarly to the normal phase, we find evidence of the bands renormalisation effect we observed in the normal phase. This phenomenon results in an enhancement of the staggered magnetisation, and allows us to connect – at Hartree-Fock level – Mott physics to the insurgence of antiferromagnetic ordering. We compare the free energy of this phase with that obtained for the normal phase, and confirm that antiferromagnetic ordering is always energetically favoured.

In the end, we study translationally invariant superconducting order. We derive the self-consistent solution for the conventional Hubbard model, and prove that superconductivity cannot be described by Hartree-Fock methods for a purely repulsive model. Upon adding the non-local pairing, we show that the new interaction fixes this problem, making superconducting order possible already at Hartree-Fock level. We derive the self-consistent solution both in singlet and triplet Cooper pairing channels, showing how Cooper pairing can be mediated at the same time by both the local repulsion and the non-local attraction, with precise internal constraints. Then we investigate numerically the singlet channel, and specifically the internal competition among the different spatial structures of the gap function. We find evidence of “symmetry mixing”, namely insurgence of a superconducting gap which is neither symmetric nor anti-symmetric under rotations of $\pi/2$ on the plane. Our data show that symmetry mixing is clearly connected to the insurgence of nematicity, namely the breaking of planar rotational symmetry. The latter is manifest in the fact that bands renormalisation, which leads also to Mott localisation, becomes anisotropic, producing a difference in electronic kinetics in directions x and y . Comparing the free energy of this phase with those of the previous two, we see that at finite doping the superconducting phase is always energetically favoured with respect to the normal phase; at null doping, instead, we identify a phase boundary among superconductivity and antiferromagnetism.

To conclude, we claim that inserting a non-local attraction in the conventional Hubbard model, various physical phenomena ranging from Mott localisation to nematic superconductivity can be addressed. Our analysis is carried out with a rather simple numerical scheme, therefore we propose an outlook for more advanced simulations.